

Stock Price Trend Prediction using LSTM

Abstract: This project predicts stock price trends using Long Short-Term Memory (LSTM) deep learning models. Historical stock market data is collected using the Yahoo Finance API. The system analyzes previous price trends and generates predictions along with technical indicators such as Moving Average and RSI.

Introduction: Stock market prediction is an important application of Artificial Intelligence and Deep Learning. This project uses Python, TensorFlow, Keras, Pandas, and Streamlit to build an interactive dashboard that visualizes stock trends and prediction graphs.

Tools Used:

- Python
- TensorFlow & Keras
- Pandas & NumPy
- Matplotlib
- yFinance API
- Streamlit

Steps Involved:

1. Collect stock data using yFinance API.
2. Preprocess and normalize the dataset.
3. Build and train the LSTM model.
4. Predict future stock prices.
5. Generate graphs for closing price, Moving Average, and RSI.
6. Deploy the dashboard using Streamlit.

Conclusion: The project successfully predicts stock trends using LSTM deep learning models and visualizes technical indicators through an interactive dashboard. It demonstrates the practical use of AI in financial data analysis.





